					Course Code:A2610001
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. I Semester MODEL QUESTION PAPER			
		ENGLISH for TECHNICAL COMMUNICATION			
		(Common to All Branches)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		Assume suitable data if necessary			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	Identify any four expressions for asking for permissions.	**1**	**2**	**2**
	b).	Complete the sentences with the appropriate words as directed in the brackets. i. The students are interested ___ learning new technologies. (Preposition) ii. The success of _____ project depends on effective teamwork. (Article)	**2**	**3**	**2**
	c).	Construct one example sentence for any two of the following idiomatic expressions. i. A piece of cake ii. Break the ice iii. Once in a blue moon iv. Spill the beans	**3**	**3**	**2**
	d).	Write about the use of skimming technique in reading the texts.	**4**	**3**	**2**
	e).	Dos and don'ts of social media messaging.	**5**	**3**	**2**
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	Develop an essay on the strategies that can help a person sound smarter, more confident, and articulate when communicating.	**1**	**3**	**5**
	b).	Write a dialogue between a student and a librarian about borrowing a reference book and renewing the library membership.	**1**	**3**	**5**
		OR			
3.		Construct a paragraph on "The Importance of Time Management" in about 100–120 words and list out the cohesive devices used in the paragraph.	**1**	**3**	**10**
		UNIT-2			
4.	**a).**	Explain how eco-conscious negotiation strategies can be applied to address an environmental issue and promote a sustainable future.	**2**	**3**	**5**

	b).	Employ appropriate verb forms to fill in the blanks. i. By next year, she ___ (finish) her degree. ii. The train ___ (arrive) at the station just now. iii. I ___ (study) English for three years. iv. When I called him, he ___ (sleep). v. Listen! Someone ___ (play) the guitar.	**2**	**3**	**5**
		OR			
5.		Apply appropriate grammatical rules and correct any **TEN** of the following sentences. i. Each of the engineers are responsible for testing the software. ii. The equipment in the laboratory were not functioning properly. iii. He discussed about the advantages of artificial intelligence. iv. The results of the experiment shows a significant improvement. v. The engineer suggested to use a different algorithm for the project. vi. If the system will fail, the backup server will be activated. vii. He is resembling his father. viii. The film show began before we reached the theatre. ix. Neither the project manager nor the team members was aware of the technical issue. x. She is an university student pursuing engineering. xi. Rahul went to school despite of having a fever. xii. I prefer chocolate milkshake than vanilla ice cream.	**2**	**3**	**10**
		UNIT-3			
6.	**a).**	Illustrate the appropriate ethical guidelines to address a challenge arising from the use of artificial intelligence or other disruptive technologies.	**3**	**3**	**5**
	b).	Construct sentences by substituting the given phrases with suitable one-word for any **FIVE** of the following. i. One who loves mankind. ii. One who cannot be defeated iii. Hobby of collecting coins. iv. Something that cannot be avoided v. A speech made without preparation vi. A person who spends money wastefully vii. A person who designs and develops software viii. A person who loves and collects	**3**	**3**	**5**
		OR			
7.	**a).**	Construct a formal letter to the Editor of a newspaper expressing your concern about the increasing use of plastic in your city. Apply appropriate letter-writing conventions and suggest possible solutions.	**3**	**3**	**10**

		UNIT-4			
8.	**a).**	Develop an essay on how innovative business practices can be applied to address challenges faced by modern startups.	**4**	**3**	**5**
	b).	Analyze the following hints to construct a formal email by applying appropriate writing techniques, tone, and structure. **Student ---> HR Manager:** New training/internship program -latest technology- useful – my career-want to be included- participate keenly-learn and implement- benefit of the company.	**4**	**3**	**5**
		OR			
9.	**a).**	Analyze the given passage and write answers to the following questions. Technology has transformed the way students learn and access information. Digital platforms, online courses, and educational applications allow learners to study at their own pace and explore subjects beyond the classroom. Engineering students, in particular, can use simulations, virtual laboratories, and collaborative tools to understand complex concepts more effectively. However, technology should complement traditional learning rather than completely replace it. Excessive dependence on digital resources may reduce face-to-face interaction and critical thinking. Therefore, students should use technology responsibly and balance digital learning with classroom discussions, practical activities, and independent study. Questions i. How has technology transformed the learning process? ii. What are the advantages of digital platforms and online courses? iii. How can engineering students benefit from simulations and virtual laboratories? iv. Why should technology not completely replace traditional learning? v. What are the possible disadvantages of excessive dependence on digital resources? vi. What does the word “complement” mean in the context of the passage? vii. Find a word from the passage that means “working together.” viii. Find a word from the passage that means “too much or more than necessary.”	**4**	**3**	**10**

		ix. What three activities does the passage recommend balancing with digital learning? x. What is the central idea of the passage?			
		UNIT-5			
10.	**a).**	Demonstrate how core values can be integrated into ethical decision-making and socially responsible leadership.	**5**	**3**	**5**
	b).	Construct the following sentences as directed in the brackets. i. She has written many novels. (Add Question Tag) ii. The report was prepared by the committee. (Convert it into Active Voice) iii. He worked hard to pass the examination. (Convert it into a Complex Sentence) iv. Very few engineering colleges in India are as reputed as IIT Madras. (Convert it into the Superlative Degree) v. Smrithi is the cleverest girl in the class. (Change it into Comparative Degree)	**5**	**3**	**5**
		OR			
11.	**a).**	Develop the content to be used in a blog on 'Social Media'.	**5**	**3**	**10**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code:A2610002

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A) **R26**

I B.Tech. I Semester MODEL QUESTION PAPER

LINEAR ALGEBRA & CALCULUS

(Common to All Branches of Engineering)

Time: 3 Hrs. **Max. Marks: 60 M**

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

5 x 2 = 10 Marks

			CO	KL	M
1.	(a)	Find the value of k such that rank of $\begin{bmatrix} 1 & 2 & 3 \\ 2 & k & 7 \\ 3 & 6 & 10 \end{bmatrix}$ is 2	1	3	2
	(b)	Find the singular values of the matrix $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$	2	3	2
	(c)	Find the value of c in [0, 4] for $f(x) = x(x-2)(x-3)$ by using Lagrange's mean value theorem.	3	3	2
	(d)	Evaluate $\underset{\substack{x\to\infty \\ y\to 2}}{Lt}\left(\frac{xy+1}{x^2+2y^2}\right)$	4	3	2
	(e)	Evaluate $\int_0^2 \int_0^4 (x^2+y^2)\,dxdy$	5	3	2

5 x 10 = 50 Marks

		UNIT-1			
2.	(a)	Determine the rank of the matrix $\begin{pmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{pmatrix}$ by reducing it to echelon form.	1	3	5
	(b)	Find the inverse of the matrix $\begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$ by using Gauss – Jordan method.	1	3	5
		OR			
3.	(a)	For what values of a and b do the equations $x + 2y + 3z = 6$, $x + 3y + 5z = 9$ and $2x + 5y + az = b$ have (i) No solution (ii) An unique solution (ii) Infinitely many solutions	1	3	5
	(b)	Solve the equations $20x + y - 2z = 17$, $3x + 20y - z = -18$and $2x - 3y + 20z = 25$ by Gauss-Seidel iteration method.	1	3	5
		UNIT-2			

4	Find the Eigen values and Eigen vectors of the matrix $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$		2	3	10
	OR				
5.	Reduce $6x^2 + 3y^2 + 3z^2 - 4xy - 2yz + 4xz$ to sum of squares by linear transformation and hence find the matrix of the transformation, Nature, Signature and index of the quadratic form.		2	3	10
		UNIT-3			
6.	(a)	Verify Rolle's theorem for $\frac{\sin x}{e^x}$ in $(0, \pi)$.	3	3	5
	(b)	Expand $f(x) = \cos x$ by Maclaurin's theorem with Lagrange's form of remainder.	3	3	5
		OR			
7.	(a)	Find the appropriate value of c used in Cauchy's Mean-Value theorem for $f(x) = \ln x$ in [1, e]	3	3	5
	(b)	Expand $\tan^{-1}x$ in powers of $(x-1)$ upto fourth degree term.	3	3	5
		UNIT-4			
8.	(a)	If $u = \cos^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, find $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$	4	3	5
	(b)	Expand $f(x,y) = e^y \log(1+x)$ in powers of x and y using Maclaurin's series.	4	3	5
		OR			
9.	(a)	If $u = \frac{yz}{x}$, $v = \frac{zx}{y}$, $w = \frac{xy}{z}$, find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$	4	3	5
	(b)	Discuss the maxima and minima of $f(x,y) = x^3y^2(1-x-y)$	4	3	5
		UNIT-5			
10.	(a)	Evaluate $\int\int xydxdy$ over the positive quadrant of the circle $x^2 + y^2 = a^2$ by changing it in to polar coordinates	5	3	5
	(b)	By the change of order of integration evaluate $\int_0^{4a}\int_{x^2/4a}^{2\sqrt{ax}} dydx$.	5	3	5
		OR			
11.		Evaluate $\iiint_V \frac{dxdydz}{x^2+y^2+z^2}$ where V is the volume of the sphere $x^2 + y^2 + z^2 = a^2$	5	3	10

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

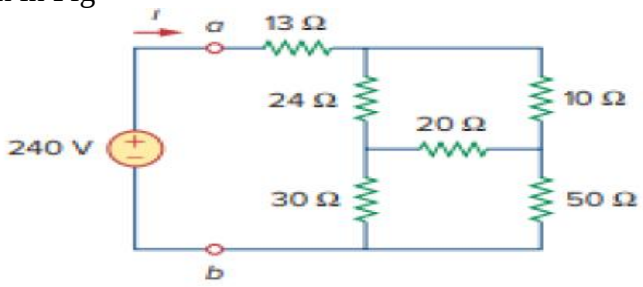
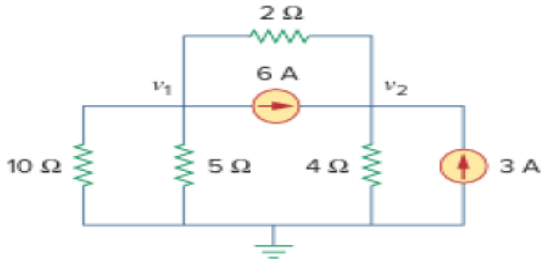
NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

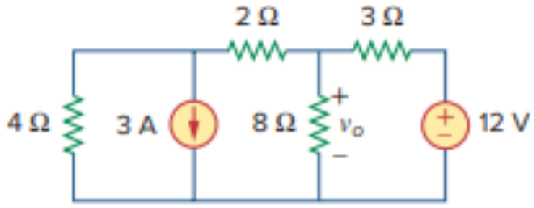
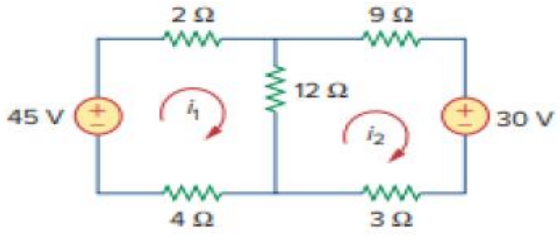
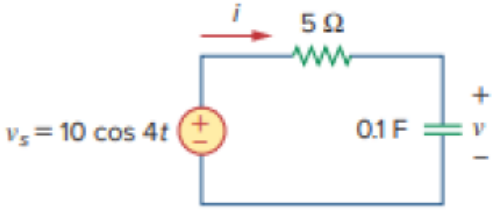
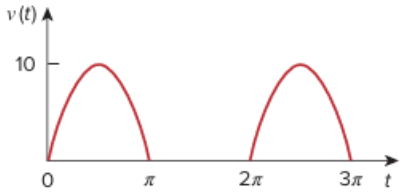
					Course Code: A2610201
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. I Semester MODEL QUESTION PAPER			
		PHYSICS FOR ELECTRICAL SYSTEMS			
		(For EEE)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE** Question from **EACH UNIT**			
		Assume suitable data if necessary			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a)**	Distinguish between ordinary light and polarized light.	1	2	2
	b)	Explain the principle of optical fiber.	2	2	2
	c)	Describe polar and non-polar dielectrics.	3	2	2
	d)	Explain the formation of energy bands in solids.	4	2	2
	e)	State the properties of matter waves.	5	2	2
					5 x 10 = 50 Marks
		UNIT-1	**CO**	**KL**	**M**
2.		Describe the formation of Newton's rings. Derive the expressions for diameters of bright and dark rings.	1	3	10
		OR			
3.	**a)**	Analyze the theory of Fraunhofer diffraction at a single slit.	1	4	10
		UNIT-2			
4.	**a)**	Describe the construction and working of He-Ne Laser.	2	3	5
	b)	Determine how population inversion can be achieved in a given laser system using an appropriate pumping mechanism.	2	3	5
		OR			
5.	**a)**	Derive the expression for Numerical aperture in terms of the acceptance angle of an optical fiber.	2	3	5
	b)	Determine the type of optical fiber based on its mode of propagation and refractive-index profile.	2	3	5
		UNIT-3			
6.	**a)**	Describe the Internal field developed in a dielectric material and establish its relationship with the applied electric field and polarization.	3	3	10
		OR			
7.	**a)**	Describe the Hysteresis curve of Ferro magnetic materials. Give the its significance.	3	3	5

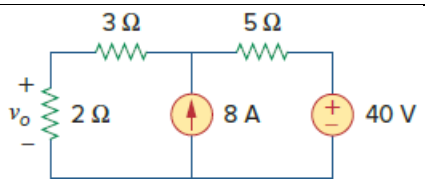
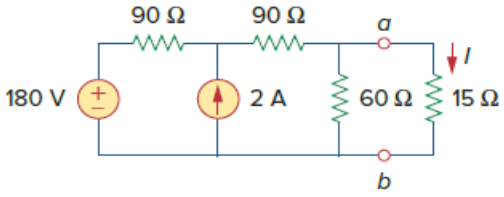
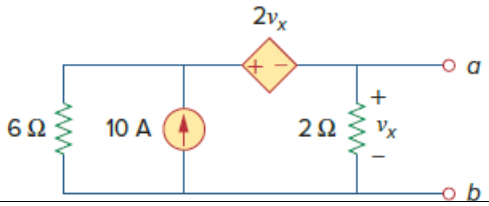
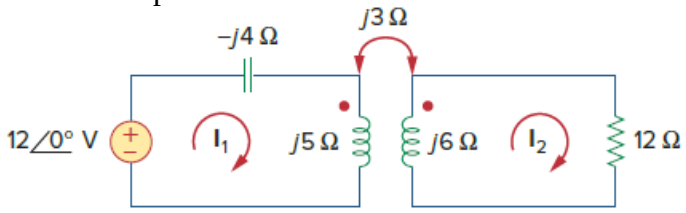
	b)	Distinguish between soft and Hard magnetic materials.	3	3	5
		UNIT-4			
8.	**a)**	Derive the expression for electrical conductivity based on Quantum free electron theory.	4	3	5
	b)	Explain the Fermi –Dirac distribution function of electrons. Explain the effect of temperature on the distribution.	4	3	5
		OR			
9.	**a)**	Derive an expression for the concentration of electrons in conduction band of an intrinsic semiconductor.	4	3	5
	b)	Describe Hall effect and derive an expression for Hall coefficient.	4	3	5
		UNIT-5			
10.	**a)**	Derive Schrodinger's time independent wave equation.	5	3	5
	b)	Apply Schrödinger wave equation to a particle confined in a one-dimensional box and calculate the resulting energies and wave functions.	5	3	5
		OR			
11.	**a)**	Distinguish between classical bits and quantum bits.	5	3	5
	b)	Calculate the output state of a qubit after applying a Hadamard gate followed by a Pauli-X gate.	5	3	5

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: A2610202
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. I Semester MODEL QUESTION PAPER			
		BASIC ELECTRICAL CIRCUITS			
		(For EEE)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE** Question from **EACH UNIT**			
		Assume suitable data if necessary			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	State Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL).	1	1	2
	b).	Define phasor and impedance in a single-phase AC circuit.	2	1	2
	c).	State Thevenin's theorem for a linear electrical network.	3	1	2
	d).	What is resonance? Define the quality factor of a resonant circuit.	4	1	2
	e).	What is earthing? State one purpose of earthing in a domestic installation.	5	1	2
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	**Calculate** equivalent resistance Rab and find current 'i' in the circuit shown in Fig 	1	4	5
	b).	For the circuit in Figure, obtain V1 and V2 using nodal analysis. 	1	4	5
		OR			
3.	**a).**	**Calculate** Vo using source transformation for the circuit shown in Fig	1	4	5

					
	b).	**Calculate** currents i1 and i2 in the circuit shown Fig using mesh analysis. 	1	4	5
		UNIT-2			
4.	**a).**	**Calculate** v(t) and i(t) in the circuit shown in Fig. 	2	4	5
	b).	A Sine wave generator supplies a 500Hz, 10V rms signal to a 2KΩ resistor in series with a 0.1μF capacitor as shown in figure. Determine the total impedance Z, current I, phase angle and voltage across capacitor and resistor.	2	3	5
		OR			
5.	**a).**	The waveform shown in Figure, is a half-wave rectified sine wave. Find the rms value and average value. 	2	4	5
	b).	The voltage across the load is v(t)=60 cos(ωt-10) V and current through the element in the direction of voltage drop is i(t)=1.5 cos(ωt+50) A. **Calculate** (a)complex and apparent powers (b)real and reactive powers (c)load impedance.	2	3	5
		UNIT-3			
6.	**a).**	Using the superposition theorem, find v_o in the circuit of Fig.	3	3	5

					
	b).	Using Norton's theorem, find the equivalent circuit across terminals ab in the circuit of Fig. Then find I. 	3	3	5
		OR			
7.	**a).**	Find the Thevenin's equivalent circuit of the circuit in Fig. at terminals a-b. 	3	3	5
	b).	Derive the condition for maximum power transfer to load in DC circuit.	3	3	5
		UNIT-4			
8.	**a).**	Derive the expression for half power frequencies in series resonance circuit.	4	3	5
	b).	A series RLC circuit has R=10Ω, L= 0.1H and C= 10μF.Calculate resonance frequency, band width, quality factor and half power frequencies.	4	3	5
		OR			
9.	**a).**	Derive the expression for co-efficient of coupling.	4	3	5
	b).	Calculate the phasor currents I1 and I2 in the circuit of Fig. 	4	4	5
		UNIT-5			
10.	**a).**	Draw and explain operation of Digital Energy meter.	5	2	5
	b).	A house has the following loads: four 60 W lamps, two 80 W fans, one 1000 W refrigerator and one 1500 W heater. Estimate the total connected load and explain how the suitable wire gauge is selected for the installation.	5	3	5
		OR			

11.	**a).**	Sketch and Explain Pipe Earthing.	5	2	5
	b).	Illustrate the construction and operation of Miniature Circuit Breaker(MCB)	5	3	5

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code:A2620001
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. II Semester MODEL QUESTION PAPER			
		DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS			
		(Common to All Branches)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		Assume suitable data if necessary			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	Find the integrating factor of $\frac{dy}{dx} - \frac{3y}{x} = x$	**1**	**3**	**2**
	b).	Solve$(D^2 - 4D + 4)y = 0$	**2**	**3**	**2**
	c).	Obtain a P.D.E from $z = ax + by + c$	**3**	**2**	**2**
	d).	Determine $\nabla\emptyset$, where $\emptyset = \sqrt{x^2 + y^2 + z^2}$	**4**	**3**	**2**
	e).	Find div $\bar{F}$, where $\bar{F} = x\bar{\imath} + y\bar{\jmath} + z\bar{k}$	**5**	**3**	**2**
					5 x 10 =50Marks
		UNIT-1			
2.	**a).**	Solve $y \log y \, dx + (x - \log y)dy = 0$	**1**	**3**	**5**
	b).	Solve $\frac{dy}{dx} - \frac{2y}{x+1} = (x+1)^3$	**1**	**3**	**5**
		OR			
3.	**a).**	If the air is maintained at 30ºC and the temperature of the body cools from 100ºC to 70º C in 15 minutes, determine the time at which the temperature of the body will be 40º C.	**1**	**3**	**5**
	b).	Find the orthogonal trajectories of the family of curves $r^2 = a^2 Sin2\theta$	**1**	**3**	**5**
		UNIT-2			
4.	**a).**	Solve $(D-2)^2 y = 8(e^{2x} + x^2)$	**2**	**3**	**5**
	b).	Solve by the method of variation of parameters $\frac{d^2y}{dx^2} + y = \tan x$	**2**	**3**	**5**
		OR			
5.	**a).**	Solve simultaneous equations $\frac{dx}{dt} + y = \sin t$, $\frac{dy}{dt} + x = \cos t$, given that x = 2 and y=0 when t =0	**2**	**3**	**5**
	b).	The damped LCR circuit is governed by the differential equation $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{c} = 0$ where L, C, R are positive constants. Find the critical resistance.	**2**	**3**	**5**

		UNIT-3			
6.	**a).**	Form the partial differential equation by eliminating arbitrary function from $f(x^2+y^2+z^2, z^2-2xy)=0$.	**3**	**3**	**5**
	b).	Solve $x^2(y-z)p+y^2(z-x)q=z^2(x-y)$	**3**	**3**	**5**
		OR			
7.		Solve $\left(D^2-3DD'+2D'^2\right)z=e^{x+2y}+Sin(x+2y)$	**3**	**3**	**10**
		UNIT-4			
8.	**a).**	Find the directional derivative of $f = xy+yz+zx$ in the direction of vector $\bar{i}+2\bar{j}+2\bar{k}$ at the point (1,2,0).	**4**	**3**	**5**
	b).	If $\overrightarrow{F} = 3x^2\overrightarrow{\imath}+5xy^2\overrightarrow{j}+5xyz^3\vec{k}$,then find $\nabla(\nabla.\vec{F})$	**4**	**3**	**5**
		OR			
9.	**a).**	Prove that $(y^2-z^2+3yz-2x)\vec{\imath}+(3xz+2xy)\vec{j}+(3xy-2xz+2z)\vec{k}$ is solenoidal.	**4**	**3**	**5**
	b).	Find the angle between the surfaces $xy^2z=3x+z^2$ and $3x^2-y^2+2z=1$ at the point (1,–2,1)	**4**	**3**	**5**
		UNIT-5			
10.	**a).**	Find the work done by the force field $\vec{F}=(2x+y^2)\vec{\imath}+xz\vec{j}+xyz\vec{k}$ in moving a particle along a straight line from (0,0,0) to (2,1,3).	**5**	**3**	**5**
	b).	Evaluate $\oint_c\{(x^2y\,dx+x^2dy\}$ where C is a circle $x^2+y^2=a^2$ by using Green's theorem.	**5**	**3**	**5**
		OR			
11.		Verify Gauss – divergence theorem for $\vec{F}=2xz\vec{\imath}+yz\vec{j}+z^2\vec{k}$ where S is an upper half surface of the sphere $x^2+y^2+z^2=a^2$	**5**	**3**	**10**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

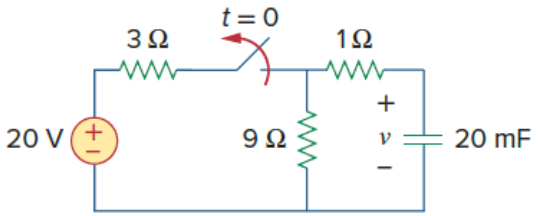
NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

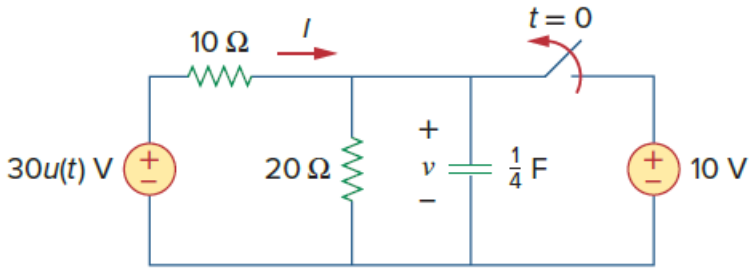
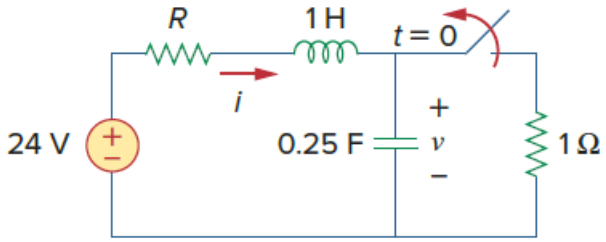
					Course Code: A2620201
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. II Semester MODEL QUESTION PAPER			
		CHEMISTRY FOR ELECTRICAL SYSTEMS			
		(For EEE)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		All Questions carry equal marks			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	**Examine why** iron corrodes faster when coupled with copper than with zinc.	**1**	**4**	**2**
	b).	Write the balanced electrochemical half-cell reaction in **Proton Exchange Membrane (PEM) fuel cell** during operation.	**2**	**3**	**2**
	c).	Mention two electronic applications of **carbon nanotubes**.	**3**	**3**	**2**
	d).	Differentiate between **bathochromic** and **hypsochromic shifts**.	**4**	**3**	**2**
	e).	How is the **threshold frequency** related to the emission of photoelectrons?	**5**	**3**	**2**
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	Describe the construction and working principle of a saturated calomel electrode with a neat labelled diagram.	**1**	**3**	**5**
	b).	Outline the principle and procedure of conductometric titration of a strong acid (HCl) against a strong base (NaOH) and sketch the sharp conductance curve.	**1**	**3**	**5**
		OR			
3.	**a).**	Describe mechanism of electrochemical corrosion by taking Rusting of Iron as an example	**1**	**3**	**5**
	b).	A steel component is exposed to differential oxygen concentrations. Apply the mechanism of differential aeration corrosion to explain its corrosion and suggest a suitable preventive technique.	**1**	**3**	**5**
		UNIT-2			
4.	**a).**	Illustrate the **construction, charging, and discharging mechanisms** of a commercial Lithium-ion battery with a neat schematic diagram	**2**	**3**	**5**
	b).	Describe the working of a bomb calorimeter and explain the corrections applied in the determination of calorific value.	**2**	**3**	**5**
		OR			

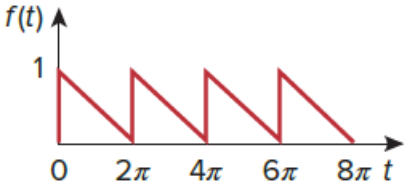
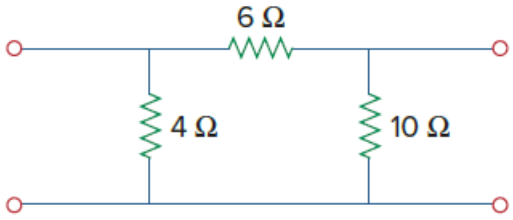
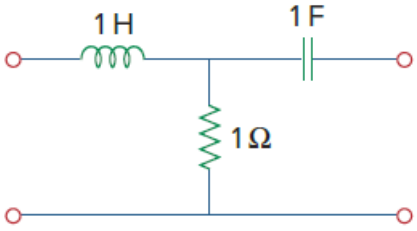
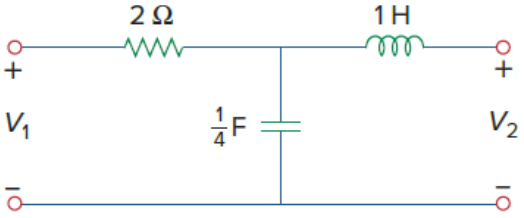
5.	**a).**	An electric vehicle requires a rechargeable battery. Apply battery characteristics such as capacity, power density and cycle life to explain their importance in selecting a suitable battery	**2**	**3**	**5**
	b).	Discuss the principle and working of Solid Oxide fuel cells and discuss its advantages?	**2**	**3**	**5**
		UNIT-3			
6.	**a).**	How does doping affect the conductivity of intrinsic semiconductor, explain with mechanism?	**3**	**3**	**5**
	b).	Apply the principle of zone refining to explain how high-purity semiconductor material can be prepared for electronic applications.	**3**	**3**	**5**
		OR			
7.	**a).**	Give preparation, properties and engineering applications of i)PVC ii) PTFE	**3**	**3**	**5**
	b).	Briefly explain the Carbon nanotubes with properties and applications?	**3**	**3**	**5**
		UNIT-4			
8.	**a).**	State the Beer-Lambert Law and derive its mathematical expression.	**4**	**3**	**5**
	b).	Why are molecules absorbed in UV-Visible region? What are the types of electronic transitions that can occur in a molecule?	**4**	**3**	**5**
		OR			
9.	**a).**	Describe the instrumentation of Infra-Red spectroscopy with a neat diagram.	**4**	**3**	**5**
	b).	Apply the working principle of MRI to explain how signals from hydrogen nuclei are used to obtain images.	**4**	**3**	**5**
		UNIT-5			
10.	**a).**	Apply Heisenberg's uncertainty principle to calculate the minimum uncertainty in momentum when the uncertainty in position is specified	**5**	**3**	**5**
	b).	What is wave function and derive the Schrodinger time independent wave equation ?	**5**	**3**	**5**
		OR			
11.	**a).**	What is a hetero diatomic molecule? Draw the energy level diagram of CO molecule and calculate its bond order?	**5**	**3**	**5**
	b).	Apply Molecular Orbital Theory to construct the π molecular orbital configuration of Benzene and calculate its bond order.	**5**	**3**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

				Course Code: A2620202	
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. II Semester MODEL QUESTION PAPER			
		ELECTRICAL NETWORK ANALYSIS			
		(For EEE)			
Time: 3 Hrs.				**Max. Marks: 60 M**	
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		All Questions carry equal marks			
				5 x 2 = 10 Marks	
			CO	**KL**	**M**
1.	**a).**	Why does the voltage across a capacitor not change instantaneously?	**1**	**2**	**2**
	b).	Draw an example of second order circuit and write its differential equation.	**2**	**2**	**2**
	c).	Determine the Laplace transform of $(t) = u(t) - u(t-2)$.	**3**	**2**	**2**
	d).	State the conditions for Reciprocity and Symmetry in terms of ABCD parameters.	**4**	**2**	**2**
	e).	Deduce the relationship between Line and Phase values of voltage and current in a balanced Delta connected system.	**5**	**2**	**2**
				5 x 10 = 50 Marks	
		UNIT-1			
2.	**a).**	Analyze the response of a series RC circuit excited by a unit step voltage (zero initial conditions). Differentiate between the natural and forced components of Vc(t), and examine how the time constant RC affects the rate at which the capacitor charges.	**1**	**4**	**5**
	b).	The switch in the circuit in Figure 1. has been closed for a long time, and it is opened at t = 0. Analyze the circuit before and after switching and find v(t) for t ≥ 0. Examine how the energy initially stored in the capacitor is shared between the 1 Ω and 9 Ω resistors during discharge.  Figure 1.	**1**	**4**	**5**
		OR			
3.	**a).**	Obtain an expression for voltage in a series R-C circuit excited by a unit step voltage. Assume zero initial conditions.	**1**	**3**	**5**

	b).	Analyze the circuit to find i and v for all time. Examine why the capacitor voltage is continuous at t = 0 while the resistor current is not, and separate v(t) into its transient and steady-state components.  Figure 2.	**1**	**4**	**5**
		UNIT-2			
4.		Derive the step response of a series RLC circuit for a DC step input. Explain the different cases of response based on the damping factor and obtain the expressions of the voltage for each case.	**2**	**4**	**10**
		OR			
5.		For the circuit shown in Figure 3. Below, determine v(t) for t > 0. R=5 Ω and analyze the response .  Figure 3.	**2**	**4**	**10**
		UNIT-3			
6.	**a).**	State and explain the Initial Value Theorem and Final Value Theorem of the Laplace transform.	**3**	**3**	**5**
	b).	Determine the inverse Laplace transform of $F(S) = \frac{2s+3}{s^2+3s+2}$	**3**	**3**	**5**
		OR			
7.	**a).**	Obtain the Laplace transform of the periodic waveform in Figure 4.	**3**	**3**	**5**

		 Figure 4.			
	b).	Find the response of RC network with sinusoidal excitation using Laplace transform method.	**3**	**4**	**5**
		UNIT-4			
8.	**a).**	Determine the y-parameters of the network shown in Figure 5.  Figure 5.	**4**	**3**	**5**
	b).	Derive the expressions for the ABCD-parameters of a two-port network in terms of its Y-parameters.	**4**	**4**	**5**
		OR			
9.	**a).**	Determine transfer impedance for the circuit shown in Figure 6.  Figure 6.	**4**	**4**	**5**
	b).	Define a transfer function? Derive the transfer function V_2/V_1 of the given Figure 7.below.  Figure 7.	**4**	**4**	**5**
		UNIT-5			

10.	**a).**	Explain the advantages of three phase system over single-phase system.	**5**	**3**	**5**
	b).	A balanced 3- phase, 3-wire 50 Hz, 100 V supply is given to a load consisting of three impedances (1+j1) **Ω**, (1+j2)**Ω**, (3+j4)**Ω** connected in star. Calculate the line and phase voltages and also currents.	**5**	**4**	**5**
		OR			
11.	**a).**	Derive voltage and current relations in Star connected Balanced load and draw the phasor diagrams.	**5**	**3**	**5**
	b).	Explain the two-wattmeter method for the measurement of active power, reactive power and power factor in a three-phase AC circuit.	**5**	**4**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: A2620203
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. II Semester MODEL QUESTION PAPER			
		COMPUTATIONAL THINKING AND PROBLEM SOLVING USING C			
		(For EEE)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		All Questions carry equal marks			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	Explain the rules for naming identifiers in C with suitable examples.	**1**	**2**	**2**
	b).	Differentiate between break and continue statements.	**2**	**2**	**2**
	c).	Explain the concept of a union in C with a suitable example.	**3**	**2**	**2**
	d).	Explain the concept of a pointer in C with a suitable example.	**4**	**2**	**2**
	e).	Explain how the #define directive is used to create a macro with an example.	**5**	**2**	**2**
					5 x 10 =50Marks
		UNIT-1			
2.	**a).**	Explain the concept of Computational Thinking and its main components with a suitable example.	**1**	**2**	**5**
	b).	Explain the different data types in C with suitable examples.	**1**	**2**	**5**
		OR			
3.	**a).**	Explain the methods used to convert a decimal number into binary, octal, and hexadecimal number systems with suitable examples.	**1**	**2**	**5**
	b).	Explain the different types of operators in C with suitable examples.	**1**	**2**	**5**
		UNIT-2			
4.	**a).**	Implement a C program using the switch statement to create a menu-driven calculator that performs addition, subtraction, multiplication, and division.	**2**	**3**	**5**
	b).	Explain the different looping statements in C with suitable examples.	**2**	**2**	**5**
		OR			
5.	**a).**	Explain the different types of if statements in C with suitable examples.	**2**	**2**	**5**
	b).	Implement a C program to sort a given array of elements in ascending order using the Bubble Sort technique.	**2**	**3**	**5**
		UNIT-3			

6.	**a).**	Explain the commonly used string handling functions in C with their syntax and suitable examples.	**3**	**2**	**5**
	b).	Implement a C program using a structure named Employee to store employee ID, name, and salary, and display the employee details.	**3**	**3**	**5**
		OR			
7.	**a).**	Implement a C program to read a string and count the number of vowels, consonants, digits, and special characters in the given string.	**3**	**3**	**5**
	b).	Explain typedef and enumerated data types in C with their syntax and suitable examples.	**3**	**2**	**5**
		UNIT-4			
8.	**a).**	Explain pointers in C, including pointer declaration, initialization, and accessing addresses and values, with suitable examples.	**4**	**2**	**5**
	b).	Illustrate the parameter passing techniques in C with suitable examples.	**4**	**3**	**5**
		OR			
9.	**a).**	Explain dynamic memory allocation in C using malloc() and calloc() with suitable examples.	**4**	**2**	**5**
	b).	Implement a C program to calculate the factorial of a given number n using recursion.	**4**	**3**	**5**
		UNIT-5			
10.	**a).**	Explain preprocessor directives and macros in C with suitable examples.	**5**	**2**	**5**
	b).	Implement a C program using file handling functions to create, store, and display student data in a file.	**5**	**3**	**5**
		OR			
11.	**a).**	Explain the commonly used file handling functions in C with their syntax and suitable examples.	**5**	**2**	**5**
	b).	Implement a C program to copy the contents of a source file into a destination file using file handling functions.	**5**	**3**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks